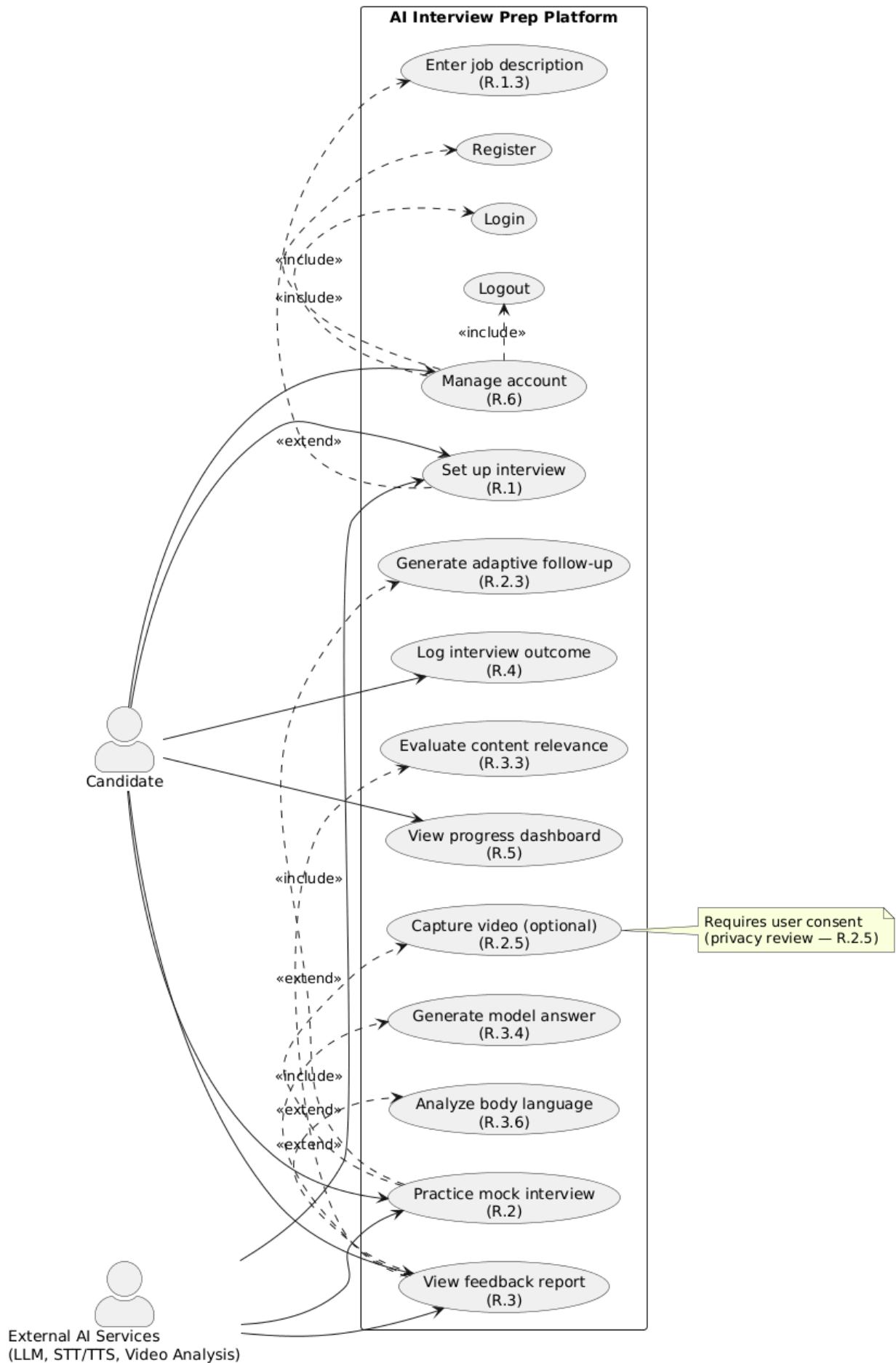
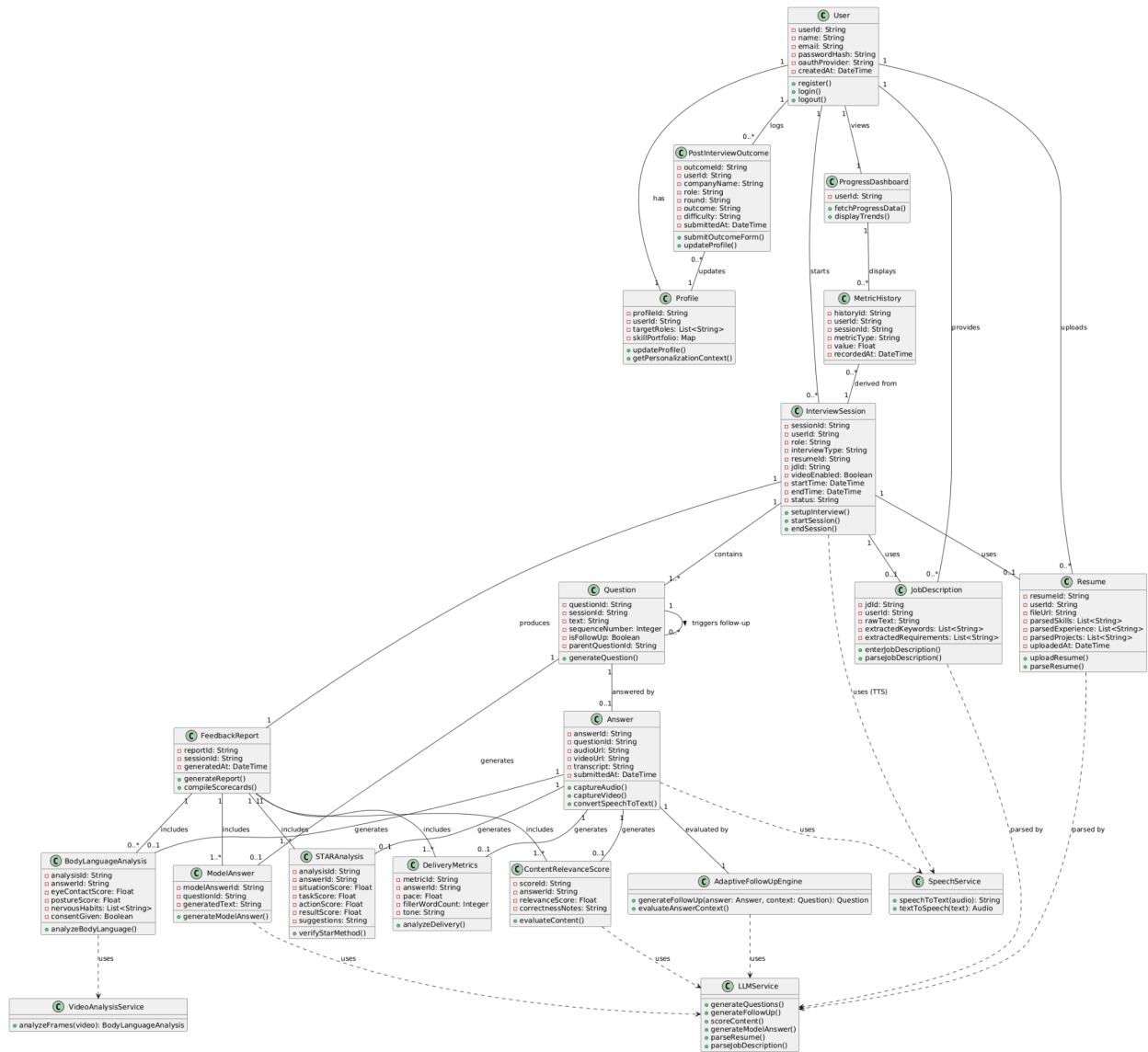


LAB-3

Use Case Diagram

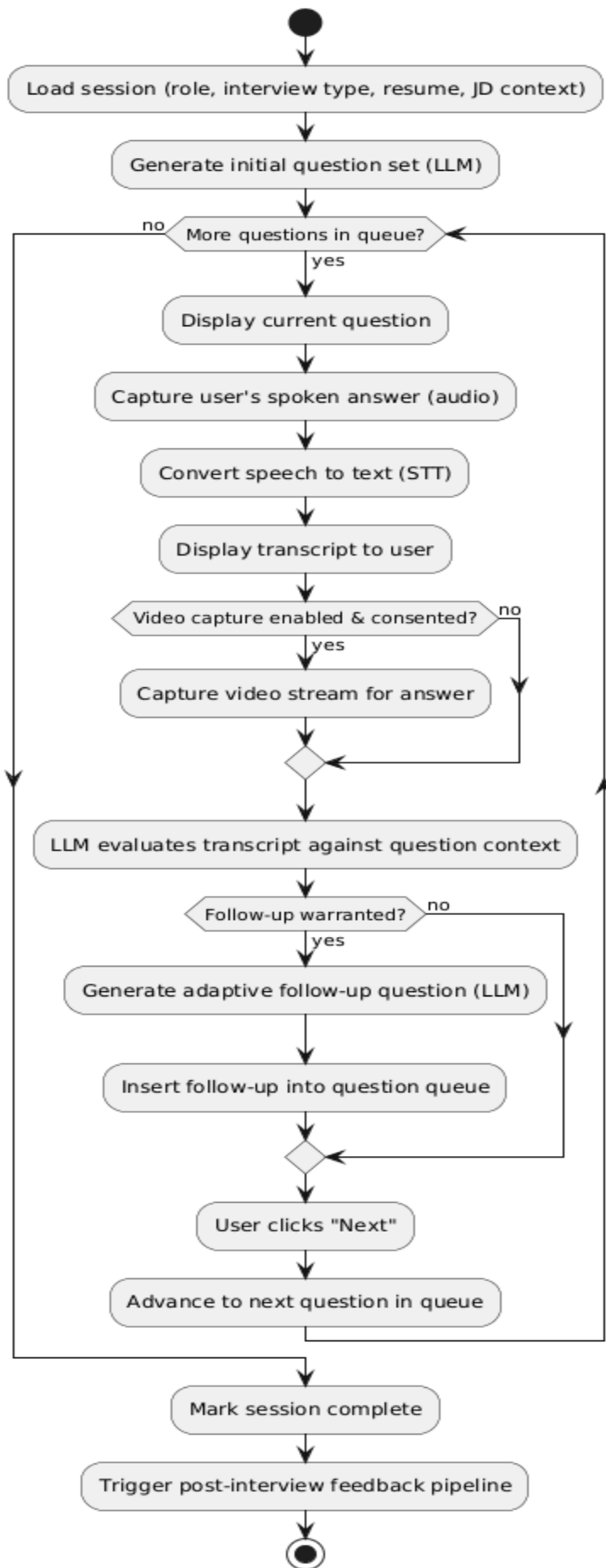


Class Diagram

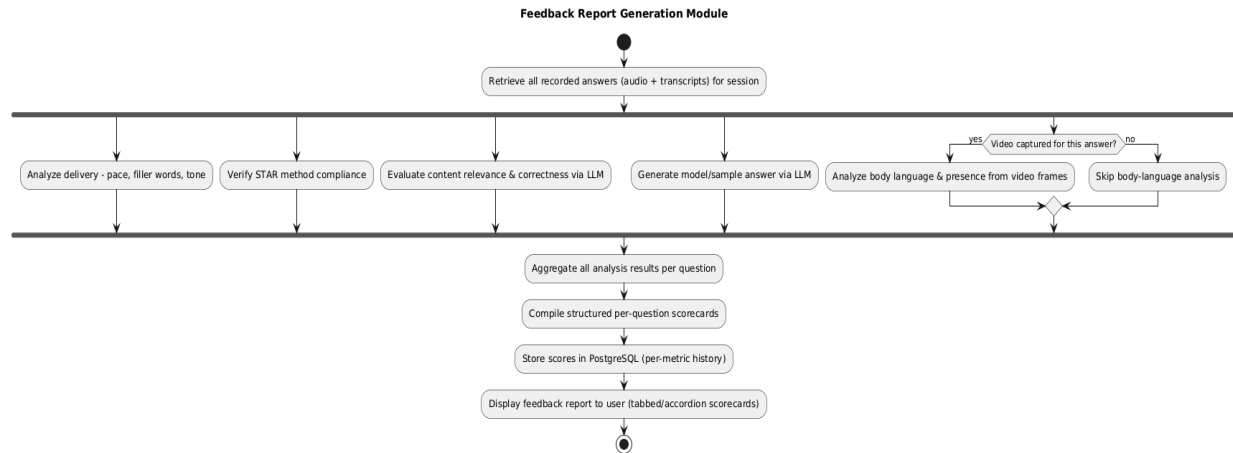


Activity Diagrams

1) Interview Session



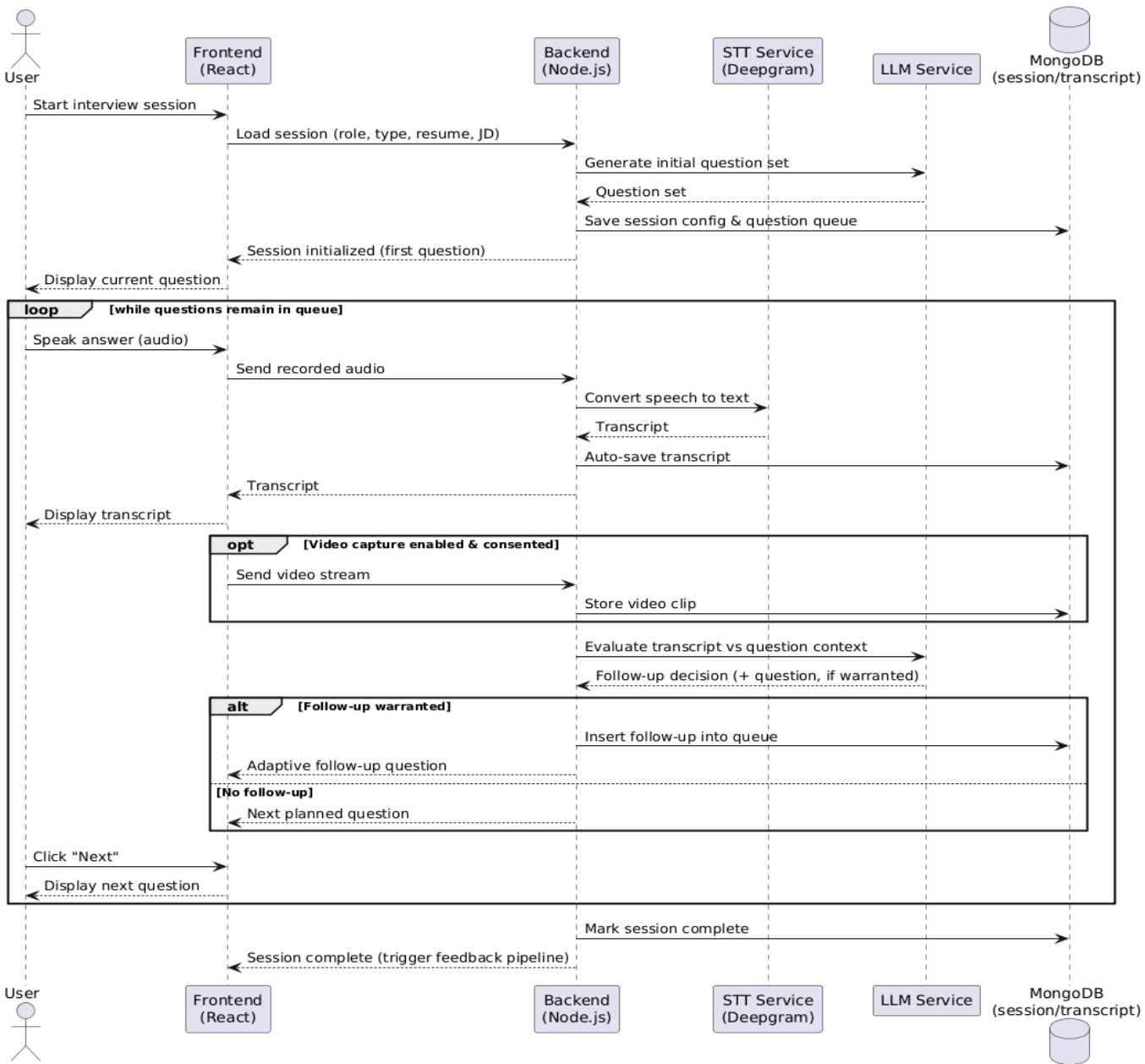
2)Feedback Report



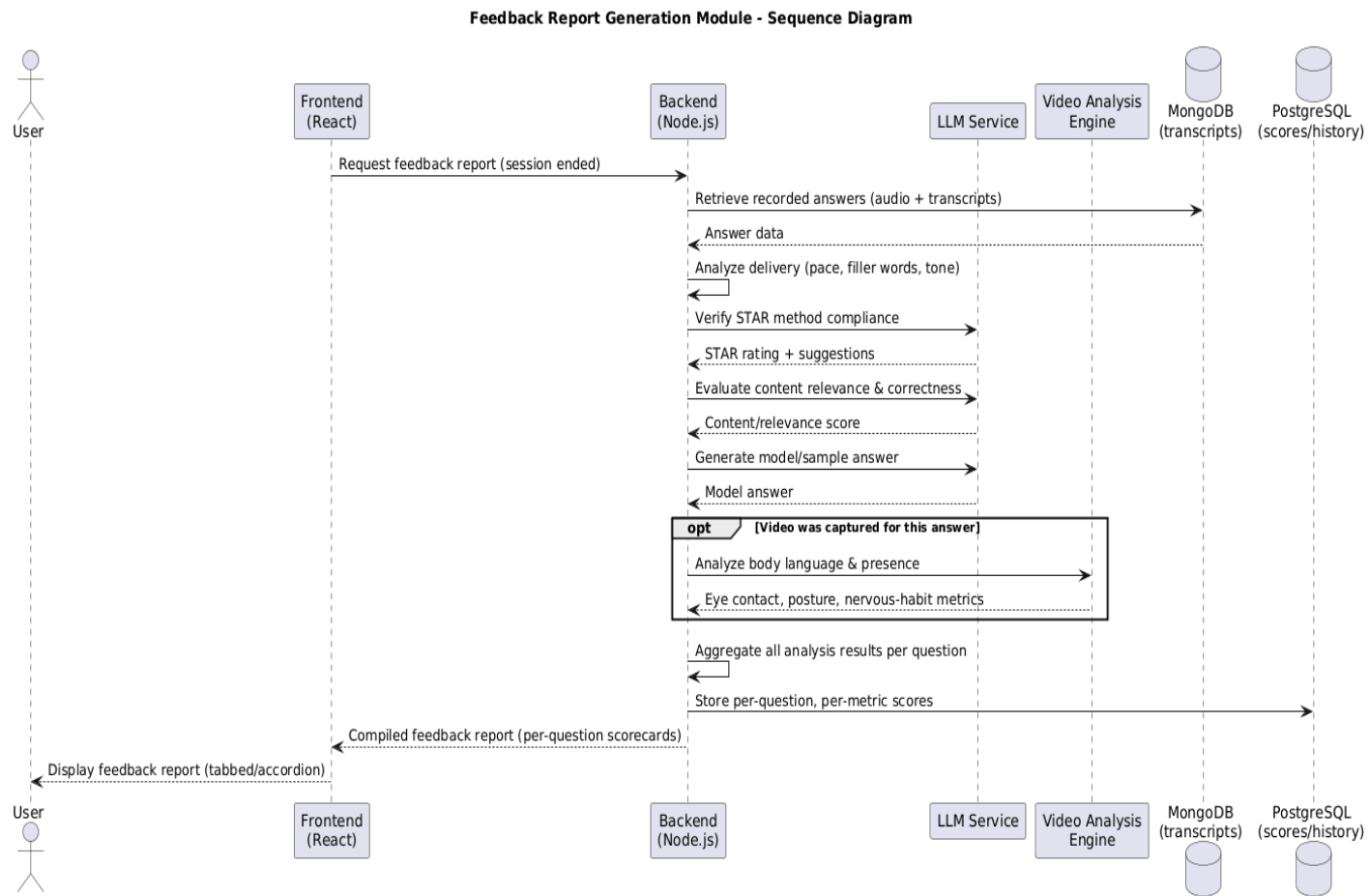
Sequence Diagrams

1)Interview Session

Interview Session Module - Sequence Diagram



2)Feedback Report



REST API Documentation

AI-Powered Interview Preparation Platform

Project	AI-Powered Interview Preparation Platform
Base URL	http://localhost:5000/api
Version	v1
Data Format	JSON (multipart/form-data for file uploads)
Authentication	Bearer token (JWT), obtained via /api/auth/login

API Summary

API	Method	Endpoint	Description
Register	POST	/api/auth/register	Create a new user account
Login	POST	/api/auth/login	Authenticate user and start session
Logout	POST	/api/auth/logout	End the current user session
Upload Resume	POST	/api/resumes	Upload and parse a resume file
Add Job Description	POST	/api/job-descriptions	Submit an optional target job description
Create Session	POST	/api/sessions	Select role/interview type and start a session
Get Session	GET	/api/sessions/:id	Retrieve session config and first question
Submit Answer	POST	/api/sessions/:id/answers	Submit a spoken answer, get transcript + next question
Upload Video Clip	POST	/api/sessions/:id/video	Attach optional video for the current answer
Complete Session	POST	/api/sessions/:id/complete	Mark session finished and trigger feedback pipeline

API	Method	Endpoint	Description
Get Feedback Report	GET	/api/sessions/:id/feedback	Retrieve the compiled per-question scorecards
Log Outcome	POST	/api/outcomes	Submit a real-world interview outcome
Get Outcomes	GET	/api/outcomes	Retrieve the user's logged outcome history
Get Dashboard Summary	GET	/api/dashboard	Retrieve summary snapshot (sessions, avg score, streak)
Get Trend Data	GET	/api/dashboard/trends	Retrieve per-metric score history for trend charts

1. Authentication (R.6)

1.1 Register

Method	POST
URL	/api/auth/register
Description	Creates a new user account.
Auth Required	None
Request Body	<pre>{"name": "Hitansh Patel", "email": "hitansh@example.com", "password": "Str0ngP@ss"}</pre>
Sample Response	<pre>{"userId": "usr_1024", "name": "Hitansh Patel", "email": "hitansh@example.com", "message": "Account created successfully"}</pre>
Status Codes	201 Created, 400 Bad Request, 500 Internal Server Error

1.2 Login

Method	POST
URL	/api/auth/login
Description	Authenticates a user with email/password (or OAuth) and returns a session token.
Auth Required	None
Request Body	<pre>{"email": "hitansh@example.com", "password": "Str0ngP@ss"}</pre>

Sample Response	<code>{"token":"eyJhbGciOiJIUzI1NiIs...","userId":"usr_1024","expiresIn":3600}</code>
Status Codes	200 OK, 400 Bad Request, 401 Unauthorized, 500 Internal Server Error

1.3 Logout

Method	POST
URL	<code>/api/auth/logout</code>
Description	Terminates the active session for the authenticated user.
Auth Required	Bearer token (required)
Request Body	None
Sample Response	<code>{"message":"Logged out successfully"}</code>
Status Codes	200 OK, 401 Unauthorized, 500 Internal Server Error

2. Interview Setup (R.1)

2.1 Upload Resume

Method	POST
URL	<code>/api/resumes</code>
Description	Uploads a resume file (PDF/DOCX) and extracts skills, experience, and projects.
Auth Required	Bearer token (required)
Request Body	multipart/form-data – field: file (PDF or DOCX, max 5 MB)
Sample Response	<code>{"resumeId":"res_501","parsed":{"skills":["React","Node.js"],"experienceYears":2},"message":"Resume uploaded and parsed"}</code>
Status Codes	201 Created, 400 Bad Request, 401 Unauthorized, 500 Internal Server Error

2.2 Add Job Description

Method	POST
URL	<code>/api/job-descriptions</code>

Description	Submits the text of a target job posting for question personalization. Optional per R.1.1.3.
Auth Required	Bearer token (required)
Request Body	<code>{"jdText":"We are seeking a Software Engineer with experience in..."}</code>
Sample Response	<code>{"jdId":"jd_318","keywords":["REST APIs","microservices","SQL"],"message":"Job description captured"}</code>
Status Codes	201 Created, 400 Bad Request, 401 Unauthorized, 500 Internal Server Error

2.3 Create Session

Method	POST
URL	<code>/api/sessions</code>
Description	Selects role and interview type, links resume/JD, and configures a new interview session.
Auth Required	Bearer token (required)
Request Body	<code>{"role":"Software Engineer","interviewType":"Technical","resumeId":"res_501","jdId":"jd_318"}</code>
Sample Response	<code>{"sessionId":"sess_9001","status":"initialized","firstQuestion":"Tell me about a challenging project you worked on."}</code>
Status Codes	201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

2.4 Get Session

Method	GET
URL	<code>/api/sessions/:id</code>
Description	Retrieves the current state of a session, including its active question.
Auth Required	Bearer token (required)
Request Body	URL Parameter: <code>id</code>
Sample Response	<code>{"sessionId":"sess_9001","role":"Software Engineer","interviewType":"Technical","status":"in_progress","currentQuestion":"..."}</code>
Status Codes	200 OK, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

3. Interview Session (Answers & Adaptive Follow-ups) (R.2)

3.1 Submit Answer

Method	POST
URL	/api/sessions/:id/answers
Description	Uploads the recorded audio answer, converts it to text (STT), evaluates it for an adaptive follow-up, and returns the next question.
Auth Required	Bearer token (required)
Request Body	multipart/form-data – field: audio (WAV/MP3); URL Parameter: id
Sample Response	<pre>{"transcript":"In my last project I led a team of...","followUp":{"question":"What was the biggest technical risk?","isAdaptive":true},"nextQuestion":null}</pre>
Status Codes	200 OK, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

3.2 Upload Video Clip

Method	POST
URL	/api/sessions/:id/video
Description	Attaches an optional video recording of the current answer, subject to user consent (R.2.4).
Auth Required	Bearer token (required)
Request Body	multipart/form-data – field: video (WEBM/MP4); URL Parameter: id
Sample Response	<pre>{"videoClipId":"vid_774","message":"Video captured for current answer"}</pre>
Status Codes	201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

3.3 Complete Session

Method	POST
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URL	/api/sessions/:id/complete
Description	Marks the session as finished once all questions are answered and triggers R.3 feedback generation.
Auth Required	Bearer token (required)
Request Body	URL Parameter: id
Sample Response	{"sessionId":"sess_9001","status":"completed","feedbackStatus":"processing"}
Status Codes	200 OK, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

4. Feedback Report (R.3)

4.1 Get Feedback Report

Method	GET
URL	/api/sessions/:id/feedback
Description	Returns the aggregated feedback report: delivery metrics, STAR rating, content/relevance score, model answer, and body-language metrics (when video was captured).
Auth Required	Bearer token (required)
Request Body	URL Parameter: id
Sample Response	{"sessionId":"sess_9001","scorecards":[{"question":"Tell me about...", "delivery":{"pace":"142 wpm", "fillerWords":4, "tone":"confident"}, "starRating":"Strong", "contentScore":8.4, "modelAnswer":"...", "bodyLanguage":{"eyeContact":"Good", "posture":"Upright"}}]}
Status Codes	200 OK, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

5. Post-Interview Outcome (R.4)

5.1 Log Outcome

Method	POST
URL	/api/outcomes

Description	Records the result of a real-world interview the user attended, to enrich future personalization.
Auth Required	Bearer token (required)
Request Body	<code>{"company":"Acme Corp","role":"Software Engineer","round":"Technical Round 2","outcome":"Rejected","difficulty":"Hard"}</code>
Sample Response	<code>{"outcomeId":"out_205","message":"Outcome submitted successfully"}</code>
Status Codes	201 Created, 400 Bad Request, 401 Unauthorized, 500 Internal Server Error

5.2 Get Outcomes

Method	GET
URL	<code>/api/outcomes</code>
Description	Retrieves the authenticated user's history of logged real-world interview outcomes.
Auth Required	Bearer token (required)
Request Body	None
Sample Response	<code>[{"outcomeId":"out_205","company":"Acme Corp","role":"Software Engineer","outcome":"Rejected"}]</code>
Status Codes	200 OK, 401 Unauthorized, 500 Internal Server Error

6. Progress Dashboard (R.5)

6.1 Get Dashboard Summary

Method	GET
URL	<code>/api/dashboard</code>
Description	Returns the card-based summary shown at the top of the dashboard.
Auth Required	Bearer token (required)
Request Body	None
Sample Response	<code>{"totalSessions":18,"averageScore":7.9,"currentStreak":5}</code>
Status Codes	200 OK, 401 Unauthorized, 500 Internal Server Error

6.2 Get Trend Data

Method	GET
URL	/api/dashboard/trends
Description	Returns per-metric score history (pace, filler words, STAR score, content score) across sessions for trend charts.
Auth Required	Bearer token (required)
Request Body	Query Parameters: metric (optional), from, to (optional date range)
Sample Response	<pre>{"metric": "contentScore", "history": [{"sessionId": "sess_8990", "date": "2026-07-20", "value": 7.1}, {"sessionId": "sess_9001", "date": "2026-07-27", "value": 8.4}]}</pre>
Status Codes	200 OK, 401 Unauthorized, 500 Internal Server Error

Status Codes

Code	Meaning
200	OK – request succeeded
201	Created – resource created successfully
400	Bad Request – missing or invalid input
401	Unauthorized – missing or invalid auth token
403	Forbidden – authenticated but not permitted
404	Not Found – resource does not exist
500	Internal Server Error – unexpected server-side failure

Validation Rules

Field	Rule
name	Required, 2–100 characters
email	Required, valid email format, unique
password	Required, minimum 8 characters, at least 1 number
role	Required (e.g., Software Engineer, Data Analyst)

Field	Rule
interviewType	Required – one of: Technical, HR, Behavioral
resume file	Required for setup, PDF or DOCX, max 5 MB
jdText	Optional, max 10,000 characters
audio file	Required per answer, WAV or MP3, max 10 MB
video file	Optional, WEBM or MP4, max 50 MB, requires consent flag
company / role / round / outcome / difficulty	All required in outcome form